Akash Bhavsar

+91 8780154490 | akashbhavsar1510@gmail.com | akashxagi | akashxagi.github.io

SUMMARY

AI Engineer with 2+ years delivering production LLM systems — hybrid RAG pipelines, multi-agent workflows, and on-premise inference stacks — across automotive, HR-tech, and compliance verticals. Expertise spanning model selection, prompt engineering, fine-tuning, and inference optimization through to production deployment, monitoring, and scaling.

SKILLS

AI/ML: LLMs, Generative AI, RAG, Agentic RAG, MCP, LoRA / QLoRA Fine-tuning, LLM Evaluation, Prompt Engineering, Machine Learning, Deep Learning, NLP
Frameworks & Tools: LangChain, LangGraph, MCP, FastAPI(SSE), PEFT, LangSmith, Opentelemetry, PyTorch, Hugging Face, scikit-learn, XGBoost, vLLM, OpenAI API, Google Gemini, Sentence Transformers, Pandas, NumPy, Pydantic
Languages: Python, SQL, Bash
Databases & Storage: PostgreSQL, pgvector, MongoDB, Redis, DuckDB, FAISS, weaviate, Qdrant, Parquet
Cloud & Infrastructure: AWS (SageMaker, EC2, RDS, S3), Docker, Nginx, GitHub Actions CI/CD, MLflow, Kubernetes

EXPERIENCE

Data Scientist, *Bacancy Technology, Ahmedabad* — Jul 2024 – Present

- Architected and shipped 7 production AI systems across automotive, HR-tech, and compliance verticals, owning the full engineering lifecycle from technical scoping and system design to deployment and post-launch optimization.
- Built reusable RAG and LangGraph agent templates adopted across enterprise client projects, reducing time-to-deployment for new AI systems by 40% and establishing internal engineering standards for LLM application architecture.

Data Science Intern, *Bacancy Data Prophets, Ahmedabad* — Jan 2024 – Jun 2024

- Built Generative AI prototypes using LLMs for enterprise clients; analyzed complex datasets with Python and SQL to drive product decisions.

PROJECTS

AI Sales Assistant SaaS for automotive dealership

- Engineered a production RAG system with SSE token streaming and hybrid retrieval (pgvector dense search + structured filter reranking) across 25+ vehicle attributes, serving natural language.
- Increased retrieval relevance by 35% (MRR@5 on held-out query set) through domain-specific embedding and ranked context injection; reduced factual hallucination rate from 22% to 8% in grounded vehicle attribute queries, validated via automated eval harness.
- Architected a 14-tool ReAct agent automating the end-to-end automotive sales cycle, eliminating 90% of manual dealer workflow steps.
- Deployed on AWS EC2 behind Nginx reverse proxy with Uvicorn workers managed by Supervisor; automated nightly inventory sync via SFTP + Cron + S3 pipeline with full RAG re-indexing — achieving zero manual ops overhead and stable uptime.

No Code AI/ML and analytics platform

- Automated end-to-end ML workflows including model training and hyperparameter tuning — cutting turnaround time by 40% using Apache PySpark and integrating MLflow Model Registry for versioning, lineage tracking, and monitoring.
- Deployed containerized local LLM inference service using vLLM with 4-bit GPTQ-quantized LLaMA-2-7B on NVIDIA CUDA, tuned to 80% GPU memory utilization — integrated into RAG pipeline, enabling session-isolated, on-premise inference as a cost-effective and secure alternative to cloud APIs for data-sensitive workloads
- Built a self-correcting NL2SQL engine with auto-retry, enabling conversational querying over structured data.

Interview Intelligence – Autonomous Transcription & Bias Detection

- Built an autonomous meeting bot that joins live video calls, records, and triggers the full AI analysis pipeline without human intervention.
- Designed a structured LLM evaluation pipeline detecting 20 cognitive bias types using chain-of-thought prompting and constrained JSON scoring; reduced post-interview analysis from 2 hours of manual review to <40 seconds automated report generation, validated against human annotator agreement.

EDUCATION

M.Sc. Big Data Analytics, *St. Xavier's College, Ahmedabad* — Aug 2022 – Jun 2024
B.Sc. Computer Science & IT, *Indus University, Ahmedabad* — Jul 2019 – Jun 2022